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Spatial Informatics

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→ Spatial : Location / Space Relationship
Geospatial : Location on Earth

- All Geospatial are Spatial but All spatial are not Geospatial

→ GIS : Geographic Information System
(DBMS of Geographic Info)

Spatial Data
Non-Spatial Data

Storage Volume
Processing Efficiency

→ Scale :

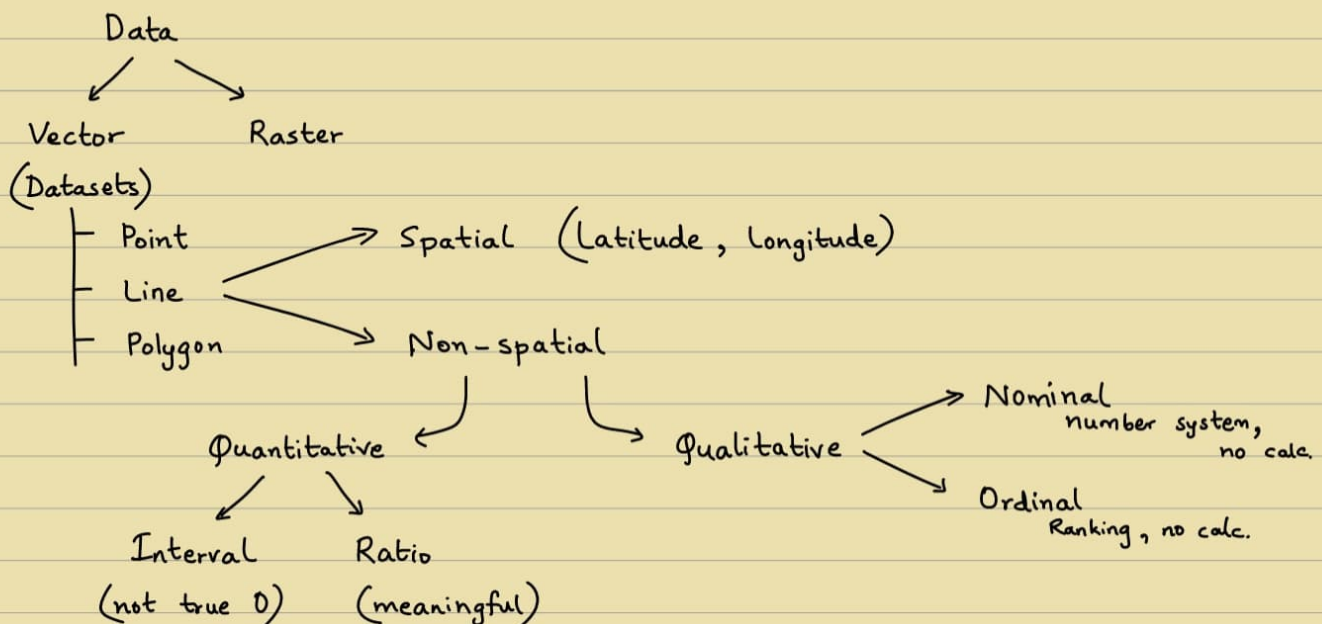
- (1) Minimize Storage
- (2) Easy to Design / Optimization
- (3) Review and Access data from database

→ Vector Data Models :

- (i) Point
- (ii) Line
- (iii) Polygon / Area

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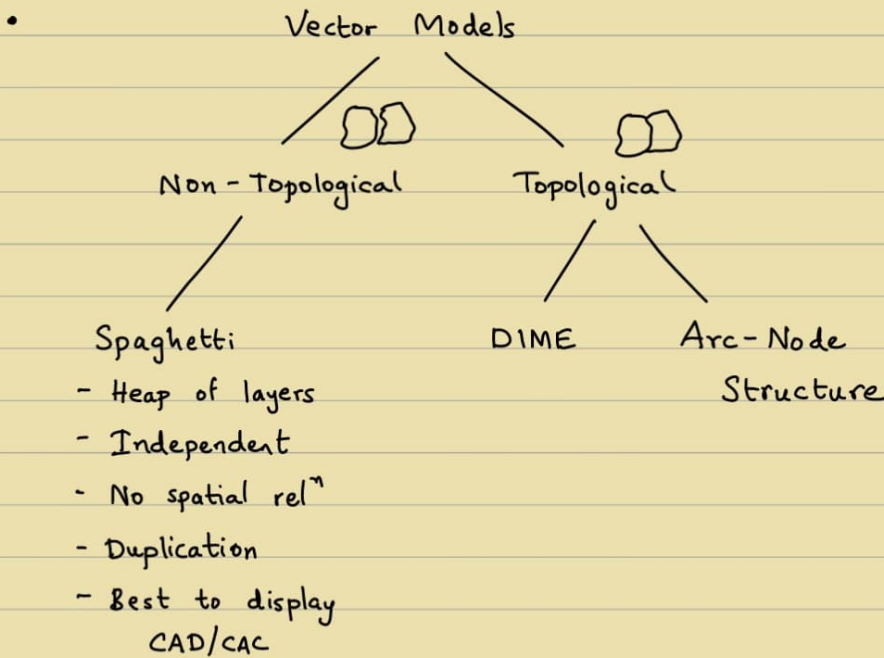
→ Geometric Entities :



- Area cannot be calculated using Points
- Nominal $\rightarrow$ Encoding
- Ratio
- Ordinal Ranking

• Chloropleth Maps :

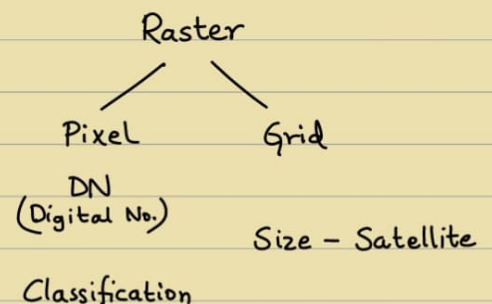
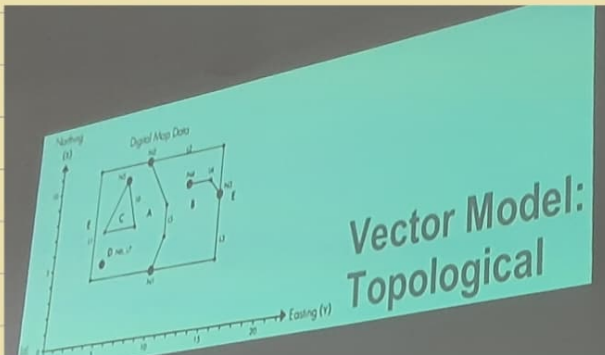
- Ranges
- Classification, then colour
- Equal Interval : Non-Technical people
- Pretty Breaks : Rounding Off
- Equal count (Quantile) : Equal no. of features
- Std. deviation
- Graduated / Proportional Symbols
- Isopleth Maps : Heat maps



DIME :

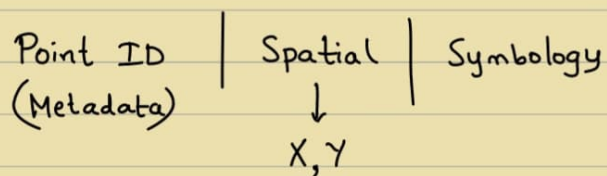
Dual Independent
Map Encoding

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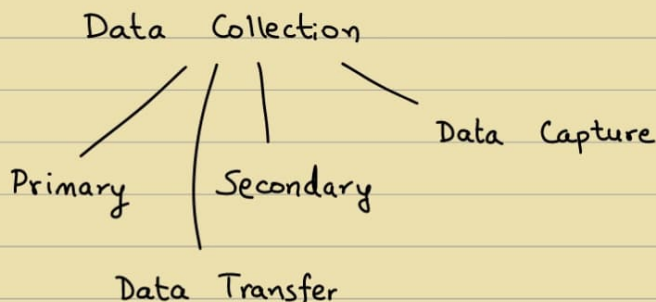


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→ GIS Layer :

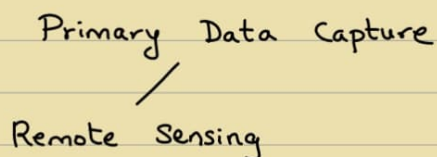


GPS → Vector



Rasterization

Vectorization



- Fast Colour Composite → Useful to study Vegetation features

Spatial Resolution : Smallest which can be viewed

Visible + IR

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- LiDAR

- Active Sensors - RADAR - Backscatter
SAR

S : Synthetic
R : Real

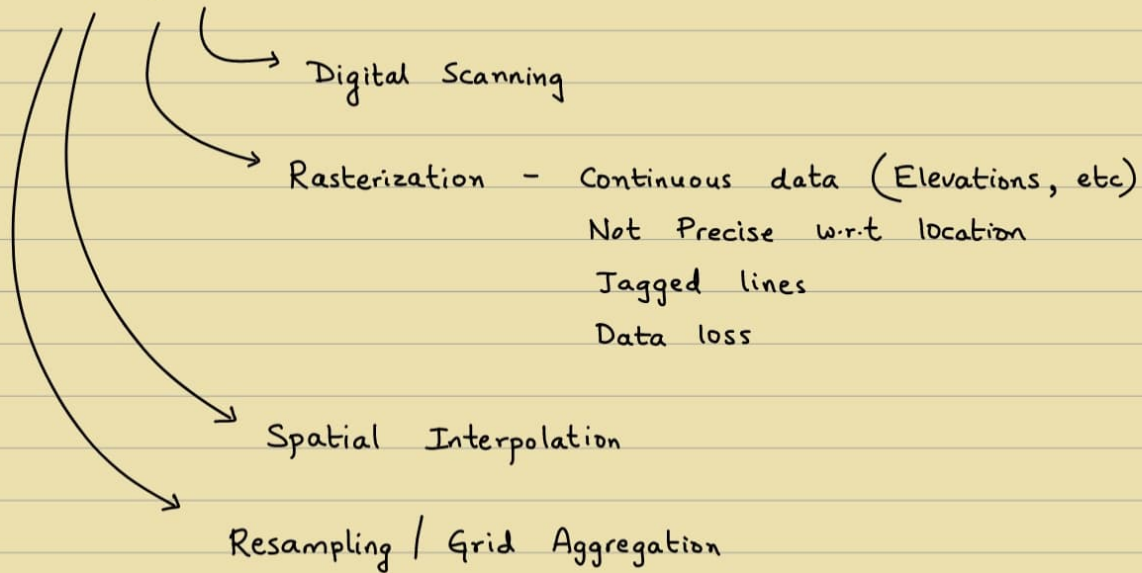
} RAR
SAR

Aperture - Antenna

RADAR - Radio Detection and Ranging

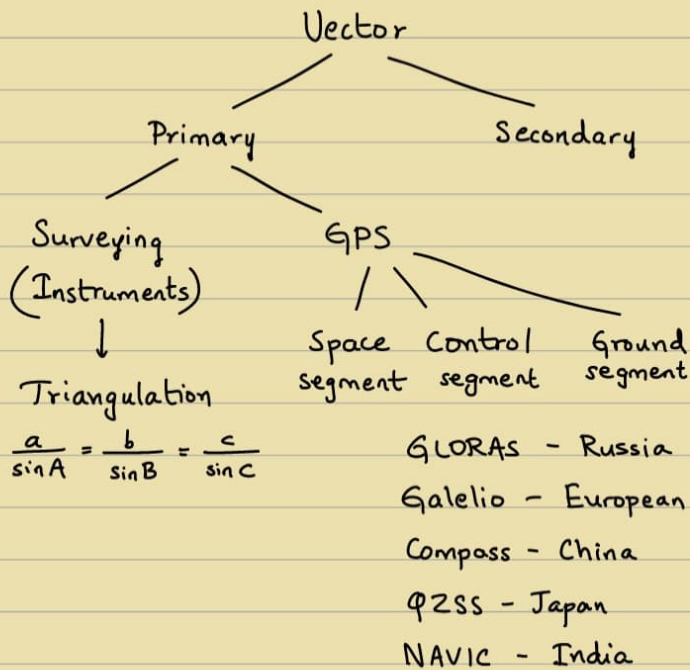
SONAR - Sound Detection and Ranging

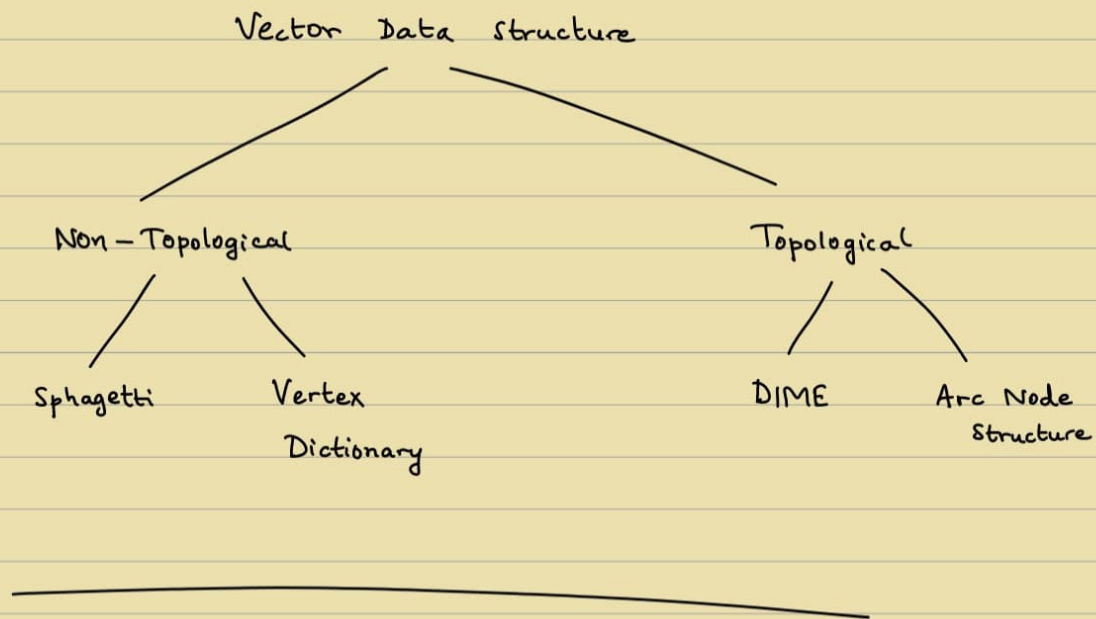
② Secondary Data Capture



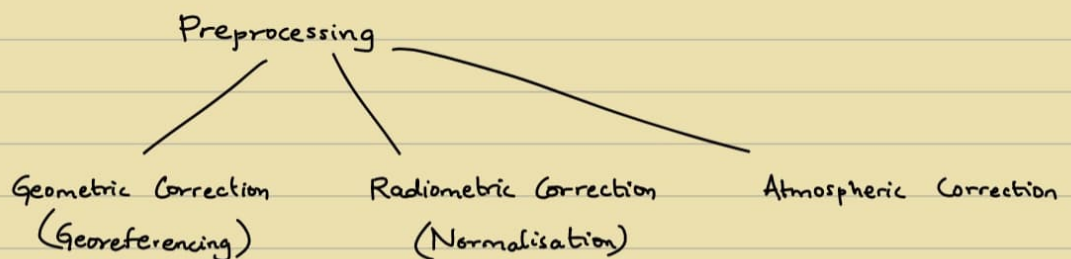
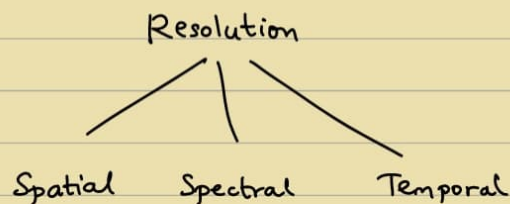
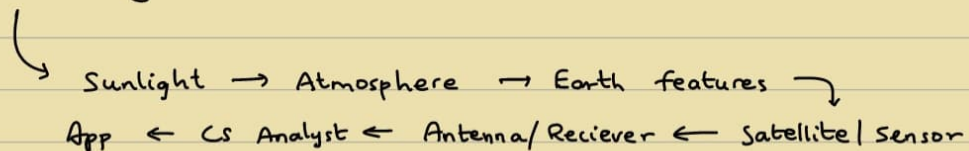
③ Data Transfer

Cloud Optimized Geotifs - COG
i.e. Geotiffs, COG, JCOG, NetCDF





Remote Sensing



LIDAR → Light Detection and Ranging

DEM : Digital Elevation Model

S → Surface

T → Terrain

TIN : Triangulated Irregular Network

RADAR

SAR → Synthetic Aperture Radar

SONAR

Mobile Mapping System

Wireless Sensor Networks

GPS :

Global Positioning System

(21 + 3) , 6 Orbital Planes

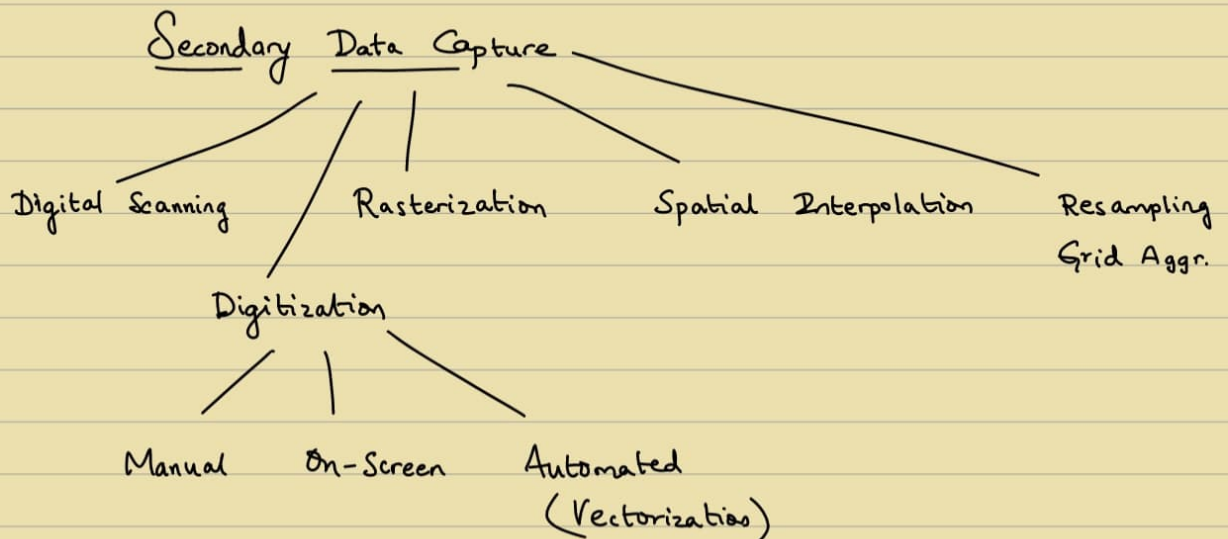
3 Satellite minimum + 1 for error correction

DGPS :

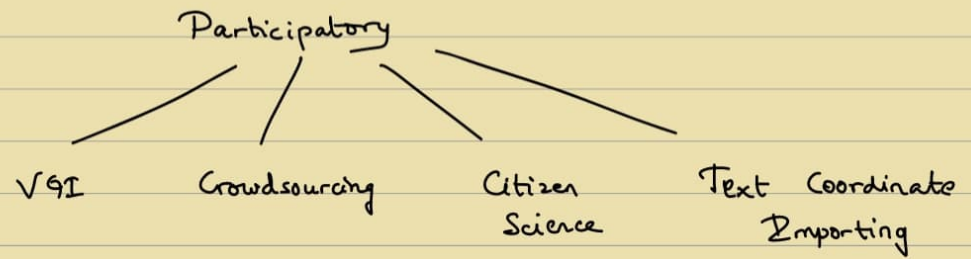
Differential GPS

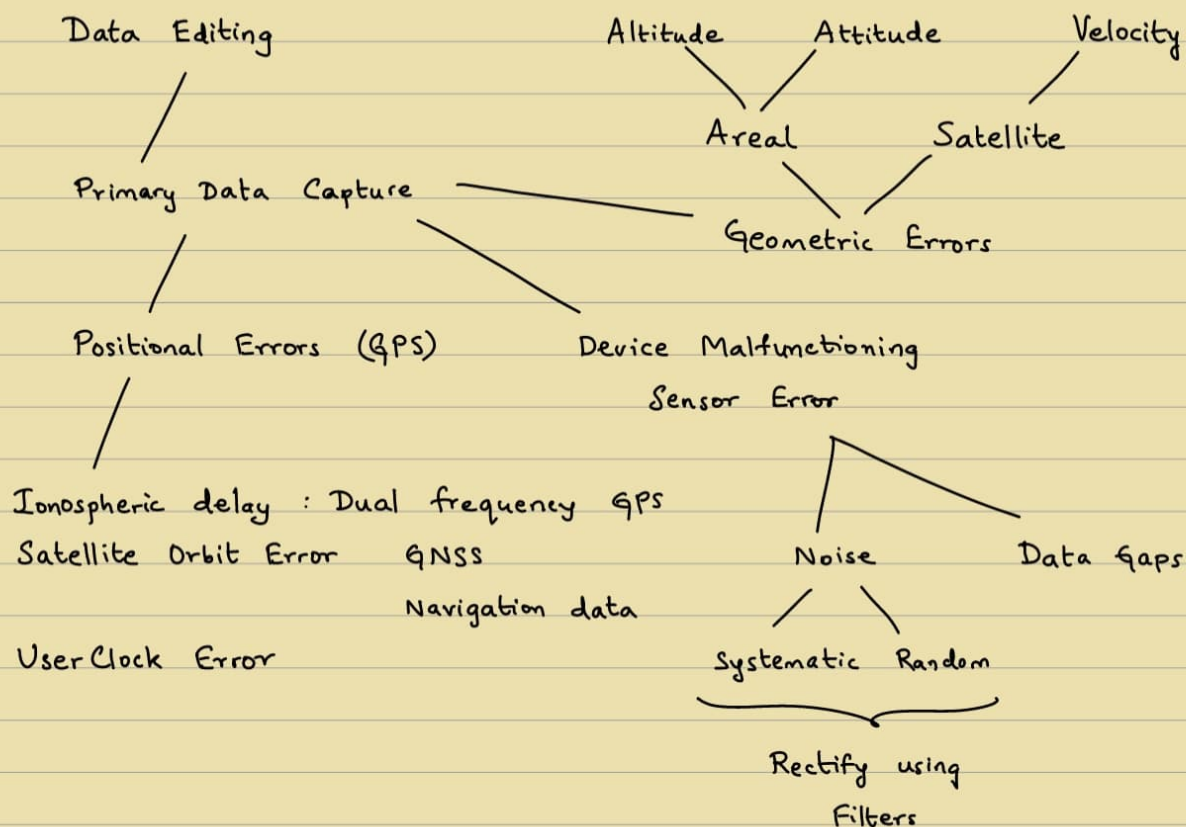
2 GPS

Ground Surveying → Theodolite



Data Transfer

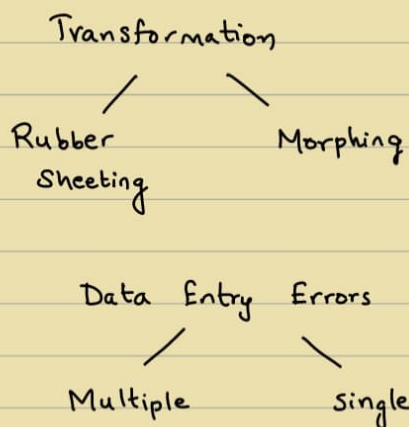




Undershoots } → Node Errors
Overshoots }

Sliver Polygons
Label Errors

Pseudo Node



→ Object Relationships

→ Relationships as Attributes

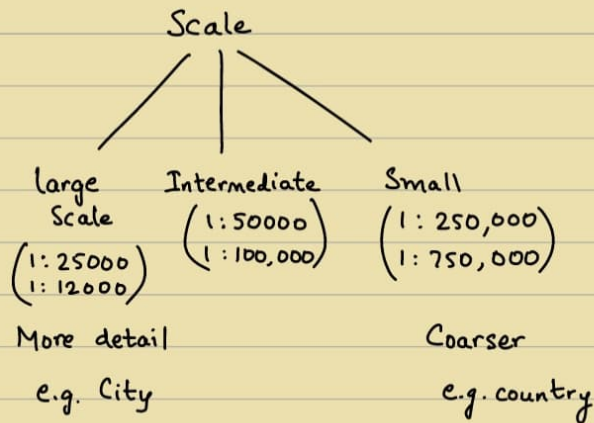
→ Topology

- Continuity
 - Connectivity
- } Shortest Path, etc

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→ Scale:

- (1) Distance & Area
- (2) Relative size of Objects that can be visualised
- (3) Neighborhood
- (4) Important areas



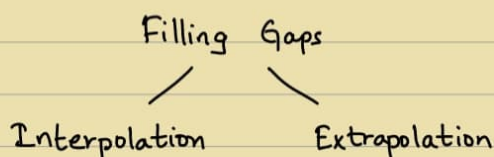
- Slope

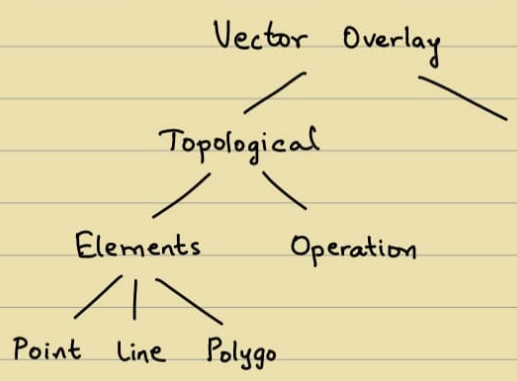
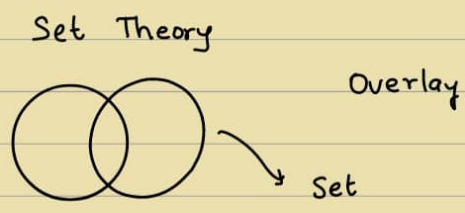
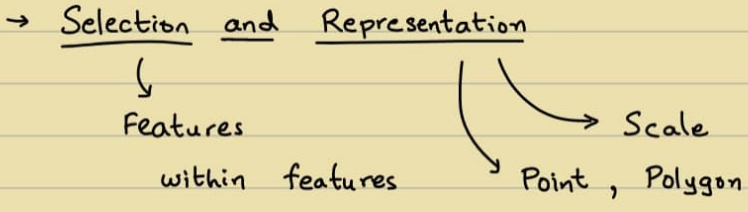
Data Integration - Errors

Crowding of features → Map Generalisation

- Weeding Out

- Spatial Inference





Topological Relations:

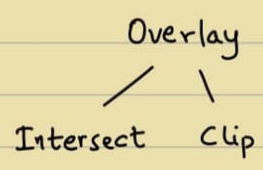
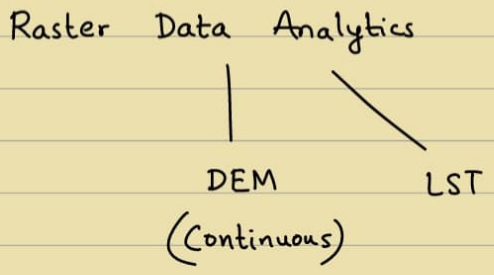
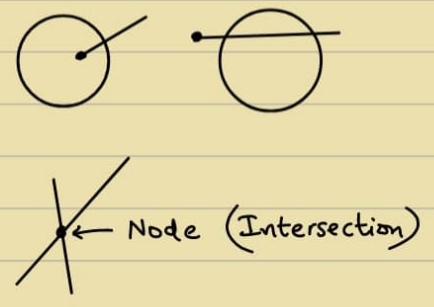
Connectivity

Adjacency / Proximity

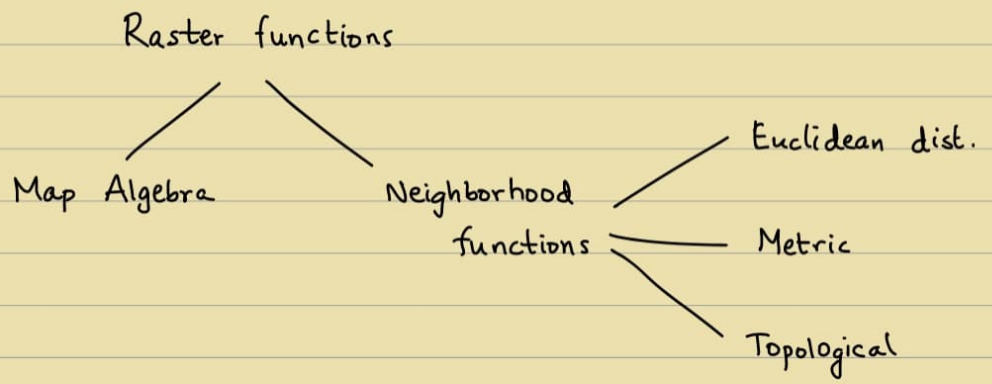
odd : Within the Polygon

even : Outside the Polygon

- Clip
- Buffer
- Intersect
- Select



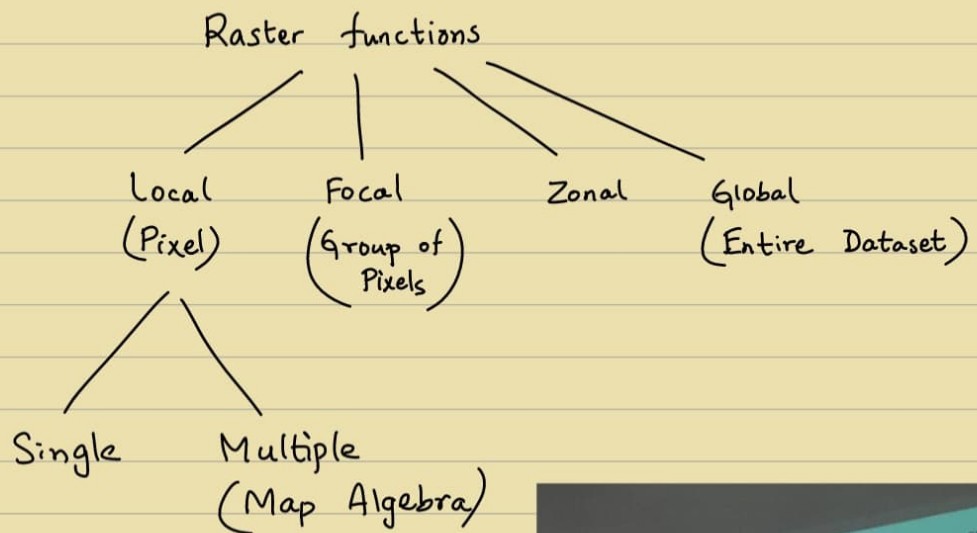
Mathematic Functions
(Likelihoods)



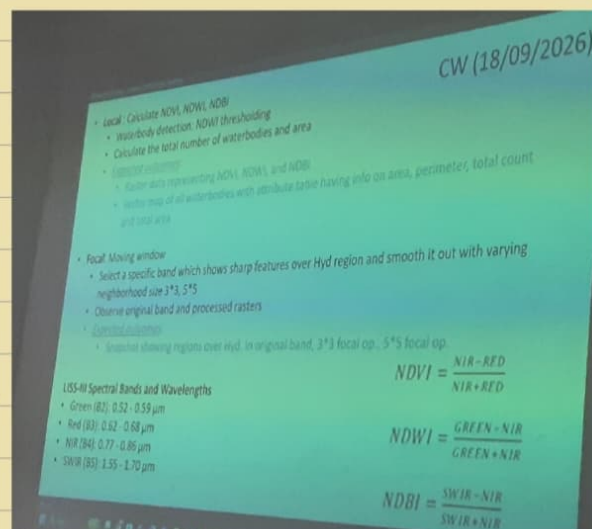
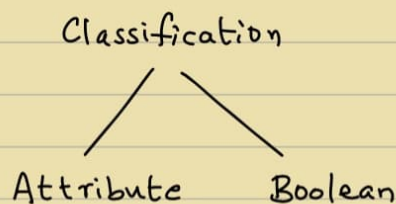
Optimal : least cost path

Adjacency
Connectivity

Enumeration Rule
Domination Rule
Contributory Rule



Reclassification - Recoding



Map Algebra

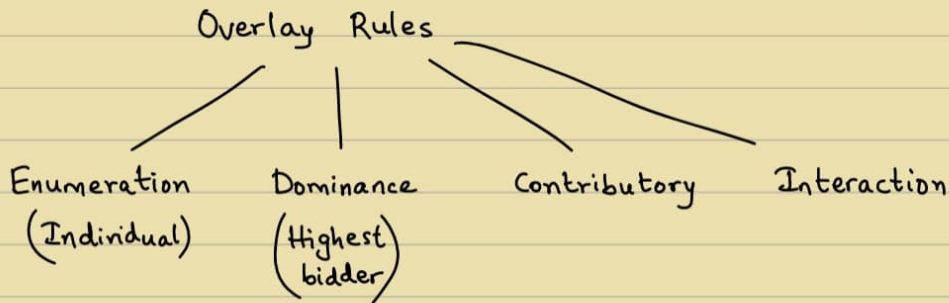
Neighbourhood

→ Adjacent Pixels

Euclidian, Metric (Distance)

Topological e.g. River, watershed

Focal → Global



Basic Operations — Focal

Local:

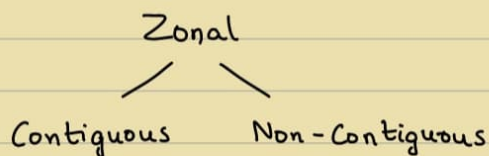
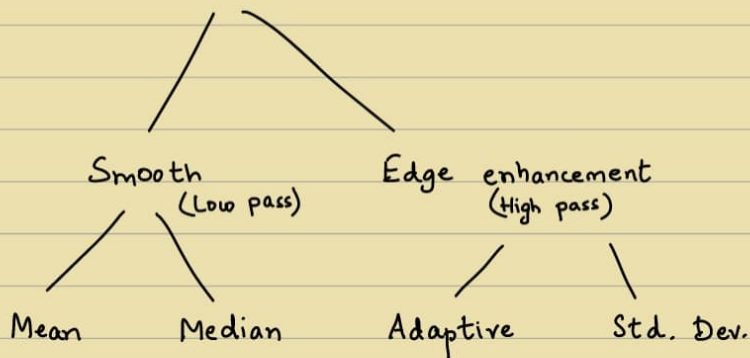
Maximise / minimize

Wedge / Pie

Reclassification

Boolean

• Spatial Convolutions / Filters



Global

- Proximity
- Buffer